ADITYA SANJAY VADALKAR

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer and 3-time hackathon winner with experience building LLM-powered systems, data platforms, and production-grade developer tools. Specialized in retrieval-augmented generation (RAG), agent orchestration, and schema-aware ML pipelines, with strong foundations in backend systems and modern frontend development. Proven ability to translate complex data and AI capabilities into reliable, user-facing products.

EDUCATION

University of Southern California — **August 2022 - May 2024**
Master's, Computer Science — *GPA: 3.67*

Savitribai Phule Pune University — **June 2018 - July 2022**
Bachelor's, Computer Engineering — *GPA: 9.42*

SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, SQL
AI/ML: LangChain, LangGraph, GPT-4, Transformers, RAG, Vector Embeddings, Semantic Search, Prompt Engineering, Fine-Tuning
Backend: Node.js, Flask, Pydantic, FastAPI, Postgres, SQLAlchemy
Frontend: React.js, Next.js, Redux, Vue.js, Material UI, Figma, Tailwind CSS
Data and Infra: Docker, AWS, Microsoft Azure, Google Cloud Platform, Serverless, ETL Pipelines

PROFESSIONAL EXPERIENCE

EasyAUM — **Remote**
Software Engineer — *April 2025 - Present*
- Built an LLM-driven market outlook system that generates grounded financial narratives from portfolio, benchmark, and risk data, replacing static templates.
- Developed a natural language to SQL interface using LangChain and GPT-4o, enabling non-technical users to query financial data during client meetings.
- Implemented schema-aware query generation and validation with SQLAlchemy and Pydantic, achieving >90% execution accuracy across 15+ query classes.
- Led a full UI/UX redesign and rebuilt the frontend in React + TypeScript, reducing user clicks by ~40% and improving engagement.

RTI International — **Remote**
Data Scientist — *July 2023 - April 2025*
- Led multiple applied ML initiatives end-to-end, including model development, evaluation, and deployment for real-world policy and healthcare use cases.
- Built production-oriented ML pipelines (Python, PostgreSQL, TF Lite) optimized for low-compute environments, balancing accuracy, latency, and hardware constraints.
- Designed data validation and transformation systems to support downstream analytics and modeling, improving reliability across multi-source datasets.
- Applied causal inference and experimental analysis to convert complex statistical results into actionable insights, directly supporting funded research programs ($200K+).

Raven — **Remote**
Founding AI Engineer — *January 2023 - October 2023*
- Owned the core AI architecture for a conversational product catalog system, designing retrieval-augmented generation pipelines over ~1M items.
- Built semantic retrieval and ranking systems using vector embeddings (FAISS, Sentence-BERT, Pinecone), optimizing relevance and response consistency.
- Developed custom agent workflows and tooling to orchestrate LLM calls, retrieval, and post-processing for deterministic outputs.
- Shipped product-facing AI features end-to-end, collaborating across frontend (React) and backend (Node.js) systems in a fast-moving startup environment.

PROJECTS & OUTSIDE EXPERIENCE

AI-Powered Audit Deficiency Extraction Pipeline
- Designed an LLM-orchestrated document processing pipeline for PCAOB reports, reducing manual extraction effort by ~90%.
- Built a modular ETL system using PyMuPDF, Pydantic, and PostgreSQL, supporting scalable vector embedding storage.
- Applied zero-shot prompt engineering to identify and categorize complex audit deficiencies.

AI-Powered Sports Highlights Generator
- Led a team of 5 developers implementing a Retrieval-Augmented Generation (RAG) system with multi-agent LLM orchestration using LangChain and OpenAI GPT-4, winning 3rd place in a multimodal hackathon
- Engineered comprehensive data pipeline for video processing and analysis, implementing speaker diarization and metadata enrichment
- Implemented an optimized hybrid retrieval system combining semantic search with metadata filtering, enabling contextually relevant content extraction from vector-embedded transcripts stored in SQLite
- *Link to project*

Step-by-Step Distillation - Llama to DialoGPT
- Utilized the QLoRA technique to finetune Llama7B on the Empathetic Dialogues dataset from Huggingface.
- Leveraged the fine-tuned model to generate rationales and curated a new empathetic dialogues (with rationales) dataset.
- Fine-tuned DialoGPT (147M) on curated dataset via step-by-step distillation, resulting in a 30% improvement.